Residual SD by distance × sequence separation
Empty: < 100 pairs cohort-wide

Layer 10 | Layer 21 | Layer 32 | Layer 47

Shared linear
d_true bin (Å)

[0,4) 3.5 | 3.1 | 4.0 | 3.7
[4,6) 2.8 3.7 3.8 3.7 3.9 4.4 | 2.7 2.7 2.6 2.6 2.7 2.9 | 2.8 3.9 3.5 3.6 3.7 3.7 | 3.5 3.4 3.3 3.2 3.4 3.4
[6,8) 3.8 3.8 3.8 4.0 4.6 | 3.0 2.6 2.6 2.7 2.9 | 3.6 3.6 3.5 3.7 3.7 | 3.2 3.1 3.1 3.1 3.2
[8,10) 3.9 3.4 3.6 3.8 4.3 | 2.4 2.4 2.6 2.7 3.0 | 3.2 3.0 3.2 3.3 3.4 | 2.3 2.2 2.4 2.3 2.4
[10,14) 3.5 3.6 3.5 3.6 4.0 | 2.4 2.6 2.6 2.7 3.0 | 2.8 2.5 2.5 2.6 2.9 | 2.0 2.0 1.8 1.8 1.9
[14,20) 3.3 3.5 3.6 3.6 3.9 | 2.2 2.4 2.4 2.6 2.9 | 2.3 2.1 2.2 2.4 2.7 | 1.8 1.6 1.6 1.6 1.7
[20,30) 3.2 3.2 3.3 3.5 | 2.5 2.7 2.8 3.1 | 2.2 2.3 2.5 2.8 | 1.6 1.7 1.7 1.7
[30,40) 3.4 3.3 3.5 3.7 | 3.1 3.2 3.1 3.3 | 3.3 2.9 3.0 3.3 | 1.6 1.9 2.0 2.1
[40,55) 3.3 3.8 3.9 | 3.8 3.9 3.9 | 3.7 4.0 4.0 | 2.4 2.7 2.9
[55,inf) 4.5 5.1 | 4.9 5.3 | 5.1 5.6 | 3.8 4.3

Per-protein linear
d_true bin (Å)

[0,4) 4.1 | 2.3 | 2.9 | 2.5
[4,6) 6.4 4.3 4.4 4.1 4.1 4.4 | 3.7 2.2 2.3 2.2 2.2 2.3 | 5.1 2.9 2.9 3.0 2.9 3.0 | 3.8 2.1 2.3 2.3 2.3 2.4
[6,8) 4.4 4.1 4.0 4.0 4.4 | 2.3 2.1 2.1 2.2 2.3 | 2.8 2.8 2.7 2.8 2.9 | 2.0 2.0 2.0 2.0 2.1
[8,10) 4.4 3.6 3.6 3.7 4.0 | 2.0 1.9 1.9 2.0 2.1 | 2.5 2.3 2.3 2.4 2.5 | 1.4 1.4 1.5 1.5 1.6
[10,14) 4.1 3.6 3.4 3.4 3.6 | 1.9 1.9 1.9 1.9 2.0 | 2.0 1.9 1.8 1.9 2.0 | 1.2 1.2 1.2 1.2 1.2
[14,20) 8.0 3.5 3.4 3.3 3.5 | 4.1 1.8 1.8 1.8 1.9 | 4.6 1.6 1.6 1.6 1.7 | 2.1 1.1 1.0 1.0 1.0
[20,30) 3.3 3.2 3.1 3.2 | 1.9 2.0 1.9 2.0 | 1.7 1.7 1.7 1.8 | 1.0 1.0 1.0 1.0
[30,40) 3.7 3.4 3.4 3.5 | 2.9 2.3 2.2 2.3 | 2.6 2.2 2.2 2.3 | 1.6 1.3 1.3 1.3
[40,55) 4.0 4.0 3.9 | 3.0 2.9 2.9 | 3.0 3.0 3.0 | 1.8 1.8 1.9
[55,inf) 5.2 5.0 | 4.1 4.0 | 4.3 4.3 | 2.7 3.0

XGBoost
d_true bin (Å)

[0,4) 1.1 | 1.2 | 1.5 | 1.4
[4,6) 1.1 1.3 1.6 1.7 2.1 2.8 | 1.0 1.5 1.4 1.4 1.5 1.7 | 0.9 1.3 1.5 1.6 1.6 1.7 | 0.5 1.5 1.1 1.2 1.2 1.2
[6,8) 1.5 1.6 1.9 2.3 3.1 | 1.6 1.5 1.5 1.7 2.0 | 1.9 1.5 1.6 1.7 1.9 | 2.0 1.2 1.1 1.2 1.2
[8,10) 1.3 1.2 1.8 2.1 2.9 | 1.2 1.1 1.4 1.5 1.8 | 1.3 1.1 1.4 1.5 1.7 | 0.8 0.7 0.9 0.9 0.9
[10,14) 1.1 1.3 1.8 2.2 3.1 | 1.0 1.2 1.3 1.4 1.7 | 1.1 1.1 1.2 1.3 1.6 | 0.6 0.6 0.6 0.7 0.7
[14,20) 1.3 1.7 2.2 2.7 3.6 | 1.2 1.4 1.4 1.6 2.0 | 1.3 1.2 1.2 1.3 1.6 | 0.9 0.6 0.6 0.6 0.7
[20,30) 2.3 2.8 3.3 4.0 | 1.7 1.9 2.1 2.4 | 1.6 1.7 1.9 2.3 | 0.7 0.8 0.8 0.9
[30,40) 2.4 3.3 3.8 4.3 | 2.1 2.6 2.8 3.3 | 2.2 2.5 2.7 3.2 | 1.0 1.3 1.4 1.4
[40,55) 3.6 4.4 4.9 | 3.6 3.9 4.4 | 3.4 4.0 4.5 | 2.5 2.4 2.5
[55,inf) 5.4 6.0 | 6.0 6.4 | 5.9 6.4 | 4.3 4.5

Sequence-gated
d_true bin (Å)

[0,4) 0.1 | 0.1 | 0.1 | 0.0
[4,6) 0.1 0.5 1.4 2.5 3.3 5.1 | 0.1 0.4 1.0 1.6 2.3 3.4 | 0.1 0.3 1.0 1.9 3.0 4.7 | 0.1 0.3 1.0 1.8 2.7 4.0
[6,8) 0.5 1.4 2.5 3.4 5.2 | 0.5 1.1 1.6 2.2 3.4 | 0.4 1.0 1.8 2.8 4.5 | 0.3 0.9 1.7 2.4 3.9
[8,10) 0.6 1.2 2.4 3.3 4.9 | 0.6 0.9 1.5 2.1 3.4 | 0.4 0.8 1.7 2.5 4.0 | 0.3 0.6 1.3 1.8 2.9
[10,14) 0.7 1.2 2.3 3.2 4.8 | 0.7 1.0 1.6 2.2 3.3 | 0.5 0.8 1.4 2.0 3.2 | 0.4 0.6 1.0 1.5 2.2
[14,20) 1.2 1.4 2.4 3.3 4.5 | 1.3 1.1 1.6 2.2 3.1 | 1.1 0.9 1.4 1.9 2.9 | 0.8 0.5 0.9 1.3 1.8
[20,30) 2.0 2.4 2.9 3.8 | 1.5 1.9 2.4 3.2 | 1.4 1.6 2.1 2.9 | 0.9 0.9 1.3 1.8
[30,40) 1.5 2.7 3.2 3.9 | 2.0 2.4 2.8 3.5 | 1.6 2.3 2.7 3.4 | 1.3 1.4 1.7 2.1
[40,55) 2.8 3.5 4.2 | 3.0 3.5 4.1 | 3.0 3.8 4.2 | 2.4 2.4 2.8
[55,inf) 4.3 5.3 | 4.6 5.4 | 5.0 5.6 | 3.6 4.2

1 2-4 5-9 10-19 20-49 50+

Sequence separation

Protein-equal mean residual SD (Å)
4.5
3.0
1.5
0.0